

CROSS-DOMAIN SPECTRAL COMPATIBILITY AND UNSUPERVISED LAND-COVER CLASSIFICATION OF VIS-NIR UAV IMAGERY IN COLOMBIAN LANDSCAPES

Juan Angel Gamez Diaz¹

Sebastián Carmona Estrada, Tutor, EAFIT University, School of Applied Sciences and Engineering, Medellín, Colombia – scarmonae@eafit.edu.co; Olga Lucía Quintero Montoya, Co-tutor, EAFIT University, School of Applied Sciences and Engineering, Medellín, Colombia –oquinte1@eafit.edu.co

ABSTRACT

Unmanned Aerial Vehicles (UAVs) equipped with Visible and Near-Infrared (VIS-NIR) sensors offer a high-resolution alternative to satellites for object recognition and land-cover analysis. This study evaluates the integration of public benchmarks and a custom Colombian UAV dataset (QUEEN-BEE) for these tasks. We performed a cross-dataset spectral similarity analysis using accumulated histograms and implemented an unsupervised K-means clustering model for land-cover classification. Our analysis revealed strong spectral alignment in the NIR band for specific vehicle classes, with Pearson correlations of 0.62 for car and 0.62 for pickup truck between VEDAI and QUEEN-BEE datasets. For land-cover clustering, the public Kaggle dataset achieved a high silhouette score of 0.57, indicating well-separated classes. Our QUEEN-BEE data, while more variable, attained a silhouette score of 0.33 using K-means with 4 clusters, which improved after geometric band alignment. These results demonstrate the significant potential of VIS-NIR imagery for cross-dataset analysis and unsupervised classification, while also highlighting the challenges posed by the inherent variability of UAV-acquired data in complex environments like Colombia.

Key words – Remote sensing; multispectral imagery; VIS-NIR bands; object recognition; land cover classification; feature extraction; histogram analysis; unsupervised learning; aerial images; VEDAI dataset; computer vision, Earth Observation, Colombia.

1. INTRODUCTION

Earth observation is fundamental for monitoring land use and land cover dynamics across multiple spatial scales [1]. Applications span agriculture, environmental risk assessment, and territorial planning, where accurate surface characterization is crucial [2]. While satellite missions provide extensive multispectral coverage, their spatial resolution often proves insufficient for fine-grained analysis in heterogeneous landscapes [3].

Unmanned Aerial Vehicles (UAVs) equipped with Visible and Near-Infrared (VIS-NIR) sensors have emerged as a powerful alternative, offering centimeter-scale resolution and flexible acquisition capabilities [4]. Their low-altitude operation enables detailed characterization of vegetation, infrastructure, and terrain features beyond conventional RGB imaging [6]. The QUEEN-BEE embedded system, developed at EAFIT University, represents such an approach for

Colombian environments.

The utility of VIS-NIR imagery hinges on understanding spectral response variations across materials, composition and land-cover types [7]. Multispectral data encodes radiation-matter interactions between electromagnetic field and surface properties, generating class-dependent reflectance patterns [8]. Effectively interpreting these patterns requires spectral space representations that reveal intrinsic structure, variability, and class separability [9].

Integrating UAV imagery with public benchmarks introduces significant challenges; geographic differences, sensor characteristics, and acquisition conditions induce spectral distribution shifts that adversely affect classification performance [10]. This creates a critical need for exploratory analysis to assess cross-dataset compatibility and identify transferable spectral features [11].

Histogram-based representations, statistical similarity metrics, and unsupervised learning provide robust frameworks for spectral structure analysis [12]. These methods elucidate class-specific intensity patterns and facilitate evaluation of whether external datasets can enhance models trained on localized data [13]. Such analysis is particularly relevant for Colombian landscapes, where spectral signatures may differ substantially from well-documented temperate regions [14].

This study addresses these challenges through three main contributions:

1. A comprehensive spectral similarity analysis between public datasets (VEDAI, Kaggle) and Colombian UAV imagery (QUEEN-BEE)
2. Assessment of cross-dataset transferability potential for object recognition and land-cover mapping
3. Evaluation of unsupervised learning performance for land-cover classification in high-variability environments

We examine spectral characteristics of VIS-NIR UAV imagery acquired in Colombia, comparing QUEEN-BEE data with public benchmarks for object recognition and land-cover classification. Our analysis identifies class-level intensity signatures, quantifies cross-dataset similarity through statistical metrics (Pearson correlation, Chi-square distance), and evaluates intrinsic separability using K-means clustering with multiple validation indices.

2. MATERIALS AND METHODS

This study follows an exploratory workflow designed to analyze the spectral structure of VIS–NIR data for object recognition and land-cover analysis. The methodology integrates dataset selection, preprocessing, statistical representation, cross-domain comparison, and unsupervised learning evaluation.

2.1. Datasets Description and Selection

Public datasets containing VIS or NIR information relevant to object recognition and land-cover mapping were surveyed. Two datasets were selected based on their spectral ranges, spatial resolution, class definitions, and acquisition conditions. A compact summary of their characteristics is presented in Table 1.

Table 1: Summary of datasets used in this study.

Attribute	VEDAI	Kaggle	L1-L2
Task	Detection	Segmentation	Clustering
Resolution	12.5 cm/px	m/px	cm/px
Image size	1024 ²	1126×1058	2592×1944
# Images	1,210	72	69-23
Bands	RGB+NIR	RGB	RGB+NIR
Spectral range	400–900 nm	400–700 nm	400–875 nm
Labels	Boxes	Masks	Masks
Classes	9 vehicles	6 land-cover	–
Location	Utah (US)	Dubai (UAE)	Colombia
Source	Aerial	Satellite	UAV

2.2. QUEEN-BEE Dataset Preprocessing Pipeline

The QUEEN-BEE embedded system outputs raw data requiring substantial preprocessing. The pipeline consisted of four sequential stages:

Step 1: Layout Extraction - RGB and NIR bands were extracted from the combined data layout generated by the QUEEN-BEE system.

Step 2: Image Division - The composite image was split vertically at the midline to separate VIS (left side) and NIR (right side) channels.

Step 3: Band Unification - The NIR band was unified with RGB bands to create 4-band multispectral TIFF images, ensuring proper spectral alignment.

Step 4: Saturation Removal - Pixel values of 255 (indicating sensor saturation) were removed from all bands to eliminate artificial intensity peaks.

The resulting preprocessed dataset, designated as L1, contained clean, multispectral patches. A subset underwent additional geometric alignment using Enhanced Correlation Coefficient (ECC) optimization with a maximum of 500 iterations and convergence threshold of 1e-4 to produce the L2 dataset with improved band-to-band registration.

2.3. Exploratory Spectral Analysis

An exploratory analysis was conducted to characterize the spectral distributions of all datasets. Accumulated intensity histograms and inter-band correlations were computed

to identify class-dependent patterns and evaluate spectral separability. A representative cumulative histogram from VEDAI is shown in Figure 1.

To visualize how labeled samples contribute to class-level spectral signatures, image patches were grouped according to their class annotations. For VEDAI, polygons defining vehicle instances delimit the pixels contributing to each class histogram. An example image and its annotation are shown in Figure 2.

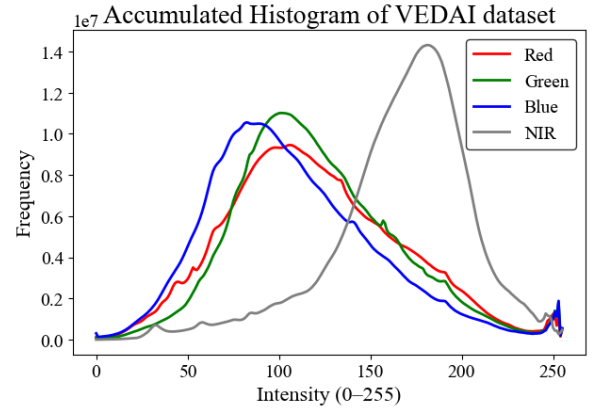


Figure 1: Accumulated histogram of VEDAI dataset.

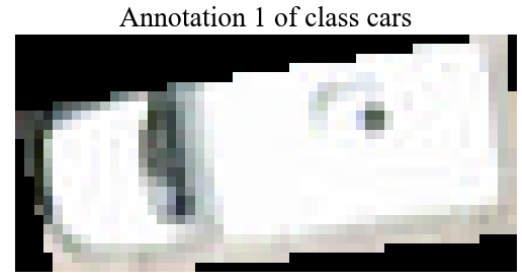


Figure 2: Example annotation of VEDAI dataset.

A similar procedure was applied to the Kaggle semantic segmentation dataset, where pixel-level masks define each land-cover category. Figures 3 and 4 illustrate a sample image and its corresponding *water* mask.



Figure 3: Example image from the Kaggle semantic segmentation dataset.

Class-level accumulated histograms were generated for all categories in both datasets.

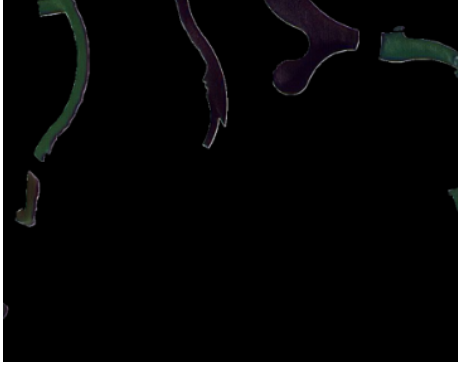


Figure 4: Mask annotation for the *water* class in the Kaggle dataset.

2.4. Cross-Domain Spectral Similarity Assessment

Cross-domain similarity was examined by comparing class-level spectral signatures of public datasets against the proprietary UAV dataset (QUEEN-BEE). Since sensors differ across platforms, the comparison focused on the NIR band to reduce spectral variability.

To quantify alignment between public and UAV-derived spectral distributions, Pearson correlation and chi-squared divergence were computed across four comparable VEDAI classes. The metrics are defined as:

$$\rho_{X,Y} = \frac{\sum_{i=1}^n (X_i - \bar{X})(Y_i - \bar{Y})}{\sqrt{\sum_{i=1}^n (X_i - \bar{X})^2} \sqrt{\sum_{i=1}^n (Y_i - \bar{Y})^2}} \quad (1)$$

where X and Y represent the two intensity distributions being compared, n is the number of histogram bins, X_i and Y_i are the frequency values in bin i , and $\bar{X}$, $\bar{Y}$ are the distribution means. The coefficient ranges from -1 to +1, indicating negative to positive linear correlation. Chi-squared is defined as:

$$\chi^2 = \sum_{i=1}^n \frac{(O_i - E_i)^2}{E_i} \quad (2)$$

where O_i represents observed frequencies from one distribution, E_i represents expected frequencies from the reference distribution, and n is the number of histogram bins. Higher values indicate greater distribution dissimilarity, with zero representing perfect alignment.

Pearson's coefficient measures linear association between intensity distributions, while the chi-squared statistic captures localized divergence between observed and expected frequencies. Together, they provide complementary evidence of cross-domain spectral compatibility.

2.5. Unsupervised Learning Evaluation

After completing the comparative analysis, the proprietary dataset was structured into three hierarchical levels: L0 contains all extracted VIS and NIR patches; L1 includes filtered and relevant samples; and L2 corresponds to geometrically aligned multispectral bands.

Unsupervised learning was applied to evaluate intrinsic separability in both public and proprietary data. K-means clustering was performed on L1, L2, and the public datasets using Min-Max normalized spectral intensities (scaled to [0,1] range). The number of clusters ($k=4$) was selected through dual criteria: visual assessment of natural land-cover groupings (vegetation, buildings, land, water) comparable to Kaggle classes, and quantitative evaluation using silhouette analysis.

Cluster quality was assessed using three complementary metrics:

$$s(i) = \frac{b(i) - a(i)}{\max\{a(i), b(i)\}} \quad (3)$$

$$S = \frac{1}{n} \sum_{i=1}^n s(i) \quad (4)$$

$$CH = \frac{\text{Tr}(B_k)}{\text{Tr}(W_k)} \cdot \frac{n - k}{k - 1} \quad (5)$$

$$DBI = \frac{1}{k} \sum_{i=1}^k \max_{j \neq i} \left(\frac{S_i + S_j}{M_{ij}} \right) \quad (6)$$

The Silhouette Score measures how similar an object is to its own cluster compared to other clusters, where $a(i)$ is the mean intra-cluster distance and $b(i)$ is the mean nearest-cluster distance for point i , with values closer to 1 indicating better-defined clusters. The Calinski-Harabasz index evaluates cluster separation using the ratio between between-cluster dispersion $\text{Tr}(B_k)$ and within-cluster dispersion $\text{Tr}(W_k)$, where higher values indicate better clustering. The Davies-Bouldin index measures the average similarity between each cluster i and its most similar counterpart j , where S_i and S_j are cluster dispersions and M_{ij} is the distance between centroids, with lower values indicating better separation.

The K-means algorithm partitions a dataset into k clusters by minimizing the total within-cluster variance:

$$J = \sum_{i=1}^k \sum_{x \in C_i} \|x - \mu_i\|^2, \quad (7)$$

where C_i is the set of data points in cluster i , μ_i is the centroid of cluster i , and $\|x - \mu_i\|^2$ is the squared Euclidean distance between point x and its centroid. The algorithm iteratively alternates between assigning data points to the nearest centroid and updating centroids until convergence. These metrics collectively describe intra-cluster cohesion, inter-cluster separation, and global structure, supporting the interpretation of VIS-NIR spectral organization across datasets.

Finally, evidence from histogram patterns, similarity metrics, and clustering behavior was integrated to characterize the spectral behavior of QUEEN-BEE data, informing the feasibility of cross-domain pattern transfer for UAV-based remote sensing in Colombia.

3. RESULTS AND DISCUSSION

3.1. Spectral Similarity Analysis Across Datasets

The exploratory analysis of spectral data revealed distinct intensity patterns for each class, both in object detection (e.g., car, truck, pickup truck) and semantic segmentation (e.g., water, land, building). Accumulated histograms show class-dependent distributions, indicating that spectral information is sufficiently separable for machine learning-based discrimination. Figure 5 compares four classes in the NIR band for VEDAI and QUEEN-BEE datasets.

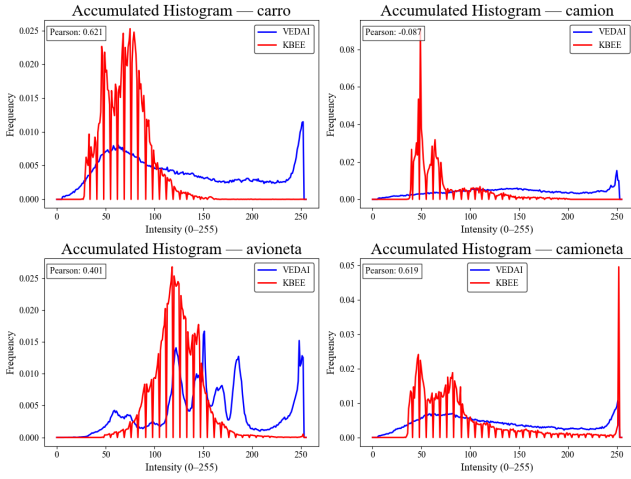


Figure 5: Accumulated NIR histograms for four classes in VEDAI and QUEEN-BEE datasets.

Table 2 summarizes the Pearson correlation and normalized Chi-square distances between VEDAI and QUEEN-BEE NIR histograms. The *car* and *pickup truck* classes demonstrate the strongest spectral alignment, with Pearson correlations of 0.621 and 0.619 respectively, coupled with the lowest Chi-square distances (0.625 and 0.569). This high correlation and low distance metric indicate that these vehicle classes share similar data representations in the spectral feature space, which is visually confirmed by their overlapping histogram patterns in Figure 5. The spectral similarity suggests these classes have consistent material properties and surface characteristics across both datasets, making them prime candidates for cross-dataset model transfer.

In contrast, the *truck* class exhibits weak correspondence (Pearson = -0.087 , Chi-square = 0.961), indicating significant spectral divergence between datasets. This discrepancy may stem from differences in truck sizes, colors, viewing angles, or material compositions between the Utah-based VEDAI data and Colombian QUEEN-BEE acquisitions. The *light aircraft* class shows intermediate similarity (Pearson = 0.401 , Chi-square = 0.777), suggesting partial but inconsistent spectral alignment.

3.2. Land-Cover Classification Performance

For land-cover classification, pixel intensities demonstrate strong correlation with specific semantic classes, illustrating through hisotgrams how K-means clustering can effectively

Table 2: Pearson correlation and normalized Chi-square distance between VEDAI and QUEEN-BEE NIR band histograms.

Class	Pearson Correlation	Normalized Chi-square
Car	0.6209	0.6245
Truck	-0.0872	0.9612
Light aircraft	0.4010	0.7773
Pickup truck	0.6186	0.5689

group intensity ranges corresponding to well-defined categories.

K-means clustering was applied to Kaggle, L1, and L2 datasets to assess intrinsic spectral separability. Figure 6 reveals that Kaggle consistently achieves higher Silhouette scores across cluster configurations, benefiting from compact, orthonormal intensity distributions characteristic of standardized satellite imagery. The QUEEN-BEE L1 dataset shows marginally better performance than L2, though both exhibit substantially lower separability compared to Kaggle.

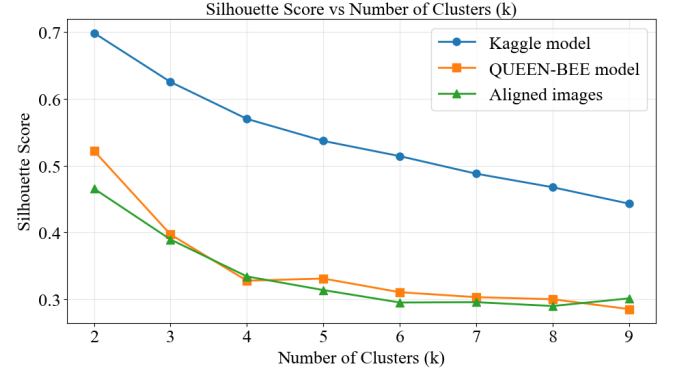


Figure 6: Silhouette scores for Kaggle, QUEEN-BEE L1, and QUEEN-BEE L2 datasets.

The significantly lower Silhouette scores in QUEEN-BEE data (0.3190 for L1, 0.3332 for L2) compared to Kaggle (0.5692) can be attributed to several factors inherent to UAV-based acquisition in Colombian environments. The greater visual dynamics present in aerial images over Colombia, compared to the more homogeneous Dubai landscape in the Kaggle dataset, increases the intra-class variability. Additionally, the non-orthorectified nature of UAV imagery introduces perspective distortions and shadow variations that expand the dynamic range of intensities, leading to more dispersed cluster distributions. The initial sensor saturation in QUEEN-BEE data, despite normalization efforts, fundamentally alters the data representation compared to Kaggle’s radiometrically calibrated imagery.

Table 3 comprehensively summarizes the clustering evaluation metrics. Kaggle achieves superior performance across all indices: highest Silhouette (0.5692), highest Calinski-Harabasz index (231,292), and lowest Davies-Bouldin index (0.5577), confirming well-separated, compact clusters. The QUEEN-BEE L2 dataset shows minor improvement over L1 in Silhouette and Davies-Bouldin metrics after geometric alignment, though the Calinski-Harabasz index decreases, suggesting that while alignment helps cluster cohesion, it doesn’t necessarily enhance overall separation structure.

Table 3: Unsupervised clustering evaluation metrics for K-means on Kaggle, L1, and L2 datasets.

Dataset	Silhouette	Calinski-Harabasz	Davies-Bouldin
L1	0.3190	69,237	1.0240
L2	0.3332	21,325	0.9331
Kaggle	0.5692	231,292	0.5577

3.3. Visual Clustering Validation

Figure 7 provides qualitative validation of the QUEEN-BEE image classification using K-means clustering. The results demonstrate that despite the spectral challenges, the unsupervised approach successfully identifies semantically meaningful clusters. Buildings are accurately classified, forming distinct structural patterns, while vegetation is clearly segmented into a separate cluster based on its characteristic spectral signature. Additionally, a well-defined cluster corresponds to road surfaces, showing the algorithm’s ability to distinguish artificial from natural land cover types.

The clustering results align with the spectral distributions observed in the histogram analysis, confirming that even with the inherent variability of UAV-acquired data in complex Colombian landscapes, K-means can capture fundamental land-cover patterns. However, the presence of mixed pixels and transitional areas between classes highlights the limitations of purely spectral-based clustering in high-variability environments.

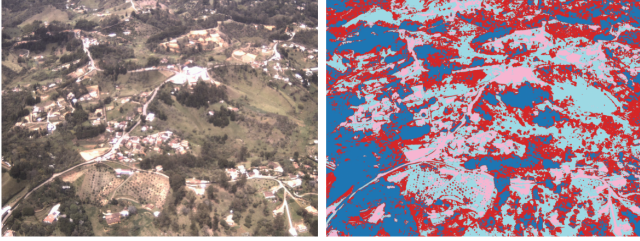


Figure 7: Example of QUEEN-BEE image classification using K-means clustering.

Overall, the results demonstrate that spectral intensity distributions enable class separation through clustering, with Kaggle providing superior separability due to standardized acquisition conditions, while QUEEN-BEE UAV data exhibits moderate but meaningful separability that is partially enhanced through geometric alignment. The cross-dataset spectral compatibility observed for specific classes like *car* and *pickup truck* suggests promising avenues for transfer learning, while the clustering performance highlights both the potential and limitations of unsupervised methods for Colombian UAV remote sensing applications.

4. CONCLUSIONS

- While n-dimensional multispectral representations provide sufficient spectral discrimination for basic land-cover classification, their effectiveness is highly dependent on environmental conditions. Colombian landscapes exhibit greater spectral variability compared to standardized datasets, necessitating adaptive approaches for reliable classification.

- The strong spectral alignment observed for specific classes (*car* and *pickup truck*) between VEDAI and QUEEN-BEE datasets (Pearson > 0.62) demonstrates that transfer learning is feasible for carefully selected object categories. However, the poor correspondence for other classes (*truck*: Pearson = -0.087) indicates that blanket transfer approaches are inadequate for Colombian conditions.
- The substantial performance gap between Kaggle (Silhouette: 0.569) and QUEEN-BEE (Silhouette: 0.333) underscores the critical importance of acquisition standardization. Sensor saturation, geometric distortions, and atmospheric variability in UAV operations introduce spectral noise that directly compromises clustering efficacy.
- Geometric alignment (L2) enhanced cluster quality over raw data (L1), confirming that basic preprocessing is beneficial. However, the modest gains (Silhouette improvement: 0.014) suggest that more sophisticated normalization techniques are needed to address the fundamental variability in UAV-acquired data.
- The consistent identification of vegetation, buildings, and roads in clustering results validates that certain land-cover types maintain stable spectral properties even in variable conditions. This enables reliable detection of key landscape elements crucial for environmental monitoring and urban planning in Colombia.
- While K-means clustering provides valuable exploratory capabilities, its performance boundaries in heterogeneous Colombian environments highlight the need for semi-supervised or feature-enhanced approaches that can accommodate the spectral diversity of tropical ecosystems.

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